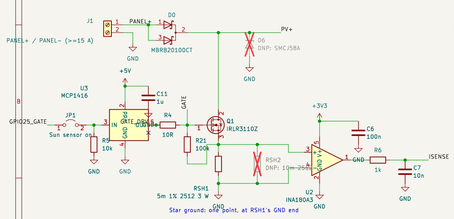
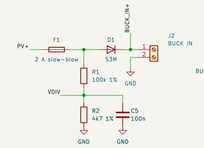
PVA carrier board v1 — schematic draft-2 (wired). Generated from hardware/CARRIER.md via parts.py. Engineer review required before manufacture.

Wired style: each capacitor and resistor is drawn beside the pin it serves. +5V, +3V3 and GND are power symbols. Net labels only where a signal leaves its block. Placement priorities: PLACEMENT.md.

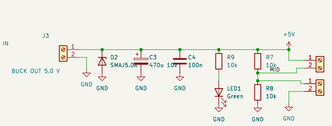
1. Panel input and sun sensor



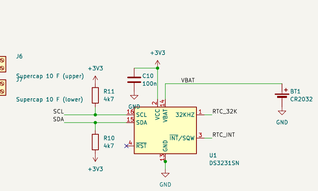
2. Board supply



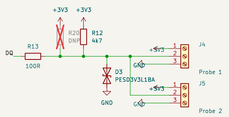
3. 5 V rail



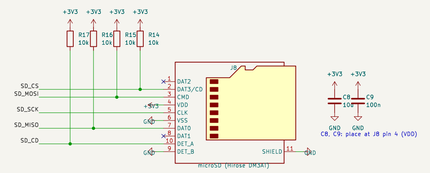
5. Clock



6. Temperature probes



7. microSD (future-proofing)



4. ESP32 socket (ELEGOO 30-pin, plugged in; antenna keep-out)

9. Power take-off for add-ons

10. AUX power out

8. Expansion headers and test points

Power flags (ERC)

Mechanical

Current paths

1. SUN-SENSOR LOOP: J1 -> D0 -> Q1 -> RSH1 -> GND (drawn in block 1)
   Panel short-circuit current, ~1 ms every 10 s. Panel Isc 5.40 A (label), max ~6 A in cold strong sun.
   A vertical panel peaks near 1.8 A at noon. The panel cannot supply more than its Isc.
2. BOARD SUPPLY: PV+ -> F1 (2 A) -> D1 (3 A) -> buck -> +5V. Under 0.3 A. F1 and D1 are NOT in the measured path.
3. AUX (optional): PV+ -> F4 (<= 3 A on v1; 9 A only with D5 on a heatsink) -> D5 -> J12. Never through the shunt.
J1 and J12 '>= 15 A' is the terminal and wire rating, not a measured current.

Measurement ranges

ISENSE (GPIO 35): RSH1 5 mOhm x INA180A3 gain 100 = 0.5 V/A. Range 0-6 A (pulsed); output clamps at 3.3 V (~6.6 A), harmless.
   U2 IN+ and IN- are Kelvin traces from the inner ends of RSH1's pads (drawn as separate wires from the shunt).
VDIV (GPIO 34): R1 100k / R2 4k7 = x 22.3. Voc 44.5 V -> 2.00 V; 48 V -> 2.15 V; 60 V -> 2.69 V. Firmware DIVIDER = (100 + 4.7) / 4.7.

ESP32 module and Q1 gate

The ESP32 is a complete ELEGOO dev board plugged into J14/J15: EN pull-up, buttons, USB-serial and strapping are on that board.
EN and GPIO 0, 1, 2, 3, 12, 15 are not brought out. GPIO 5 (strapping) is held high by R14.
Q1 gate: U3 drives it when +5V exists; R21 (100k, gate-source) holds Q1 off before that, at dawn.
D6 (TVS on PV+): footprint only; fitted if the v1 bench test shows turn-off overshoot. Then Q1 >= 100 V.

Draft: engineer review required before manufacture
Generated from hardware/CARRIER.md by gen_schematic.py (wired style)
photovoltaic-art / SUNDIAL
Sheet: /
File: carrier.kicad_sch
Title: PVA carrier board v1
Size: A1 | Date: 2026-10-26 | **Rev: draft-2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1